

JS Notes

14 June 2025 19:02

Variable Declaration:

let x = 123;

Keyword	Scope Type	Details	Can Redeclare in same scope	Hoisted	Initialised at Top?	Access before declaration
var	Function-scoped	Visible throughout the function it is declared in.	Yes	Yes(undefined)	Yes(with undefined)	✅ (but value is undefined)
let	Block-scoped	Visible only within the enclosing {} block.	No	Yes(TDZ)	No	❌ ReferenceError
const	Block-scoped	Same as let, but the value cannot be reassigned.	No	Yes(TDZ)	No	❌ ReferenceError

- const **cannot be reassigned**, but its **contents can be mutated** (like arrays/objects).

♦ TDZ = Temporal Dead Zone (accessing before declaration causes error)

To use variables in a sentence efficiently:-

```
console.log(`You are ${age}`);
```

Or

```
console.log("You are " + age);
```

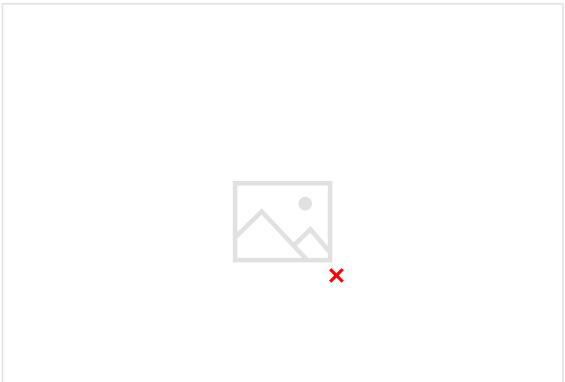
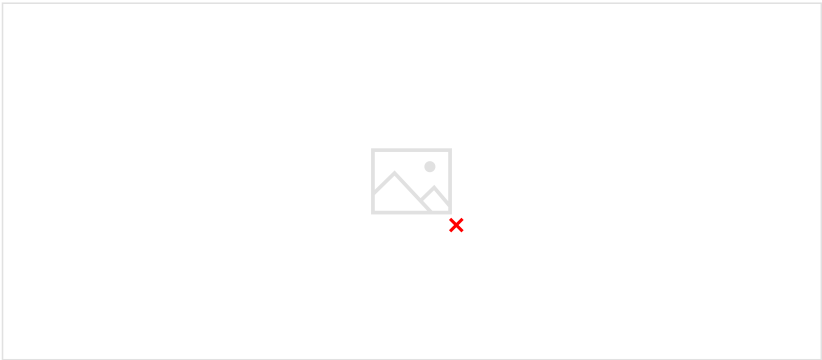
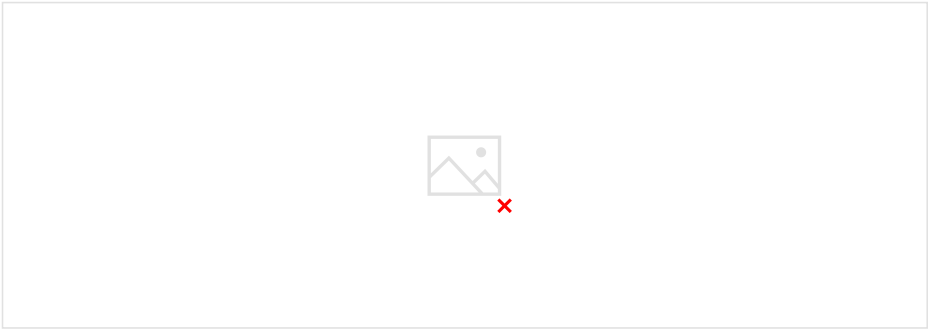
To get type of variable:

```
console.log(typeof age);
```

- Hoisting** means that JavaScript **moves declarations to the top** of the current **scope** (script or function) **before executing the code**.
- But here's the catch:
 - Only the **declarations** are hoisted.
 - Initializations (assignments)** are **not** hoisted.

```
// var old js mein tha
// var function scoped hota hai => var apne parent fnc
// mein kahi bhi use ho sakta hai
// var adds itself to the window object
//
// let const new js mein hai
// let const braces scoped hota hai
// let const doesn't adds
```







Mathematical Operations:

+ - * / ** % -> Binary
+=, -=, *=, **=, /=, %= -> Assignment
++, -- -> Unary



To manipulate data from an input, we can do:

```
InputElement.value = "value";  
Variable = InputElement.value;
```



- `.textContent` => to modify textcontent of an element
- `.getElementById("");` => To get an elmt by its ID.

Type Conversion

For conversion from one data type to another:

```
Str = Number(Str);
```

```
Num=String(Num);
```

```
Num=Boolean(num)
```

- To get type of variable ->
`console.log(typeof variableName)`

- NaN => Not a Number



- Undefined vs not-defined-
- Undefined-> existent ha but koi value nhi h ya pata nhi h
- Not-defined-> existent hi nhi h

- **Math Library Functions:**

Math.PI

Math.E (Euler Number)

Math.floor (rounds down)

Math.ceil (rounds up)

Math.round

Math.trunc (removes any decimal part)

Math.pow(x,y) => x-base, y-power

Math.sqrt

Math.log() => natural log

Math.cos(x)

Math.sin(x)

Math.tan(x)

Math.abs(x) -> Returns absolute value of number

Math.sign(x) => gives sign of the number

Math.max(x,y,z) => can pass any number of arguments separated by commas, gives max out of them

Math.min(x,y,z) => can pass any number of arguments separated by commas, gives min out of them



<https://developer.mozilla.org/en-US/docs/Web/API/HTMLAudioElement/Audio>

<https://github.com/goldfire/howler.js#documentation>

- **Random Numbers**

`Math.random()` => gives any random num bw 0 and 1

To get like a specific range of random numbers =>

`Math.round(Math.random()*10)+1`

=> will give random number between 0 to 10

To get random number in a range:

e.g. random number from 50 to 100 ->



- **Conditional Statements**

`If(cond1){if cond1 met}`

`Else if(cond2){else if cond2 met}`

`Else{todo if no conditions are met}`

- Nested conditions

- **Equality Condition**

`(x==y)` -> Only checks for values being equal.

- **Strict Equality Check**

`(x===y)` -> Checks for values being equal as well as datatype being same.

- **.checked property ->**

Property that determines the checked state of checkbox or radio.

Returns true or false



- **Ternary Operator**

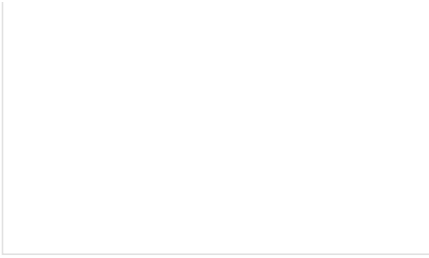
Shortcut to if-else statement, helps to assign a variable based on a condition

Condition ? codeIfTrue : codeIfFalse



- **Switch Statements**





- **String Manipulation**
 - `Str.charAt(index)`

To get character at any index of string:

- `Str.getIndex()`

To get index of a character present in a string

- `Str.length`

To get length of string

String Slicing

The `slice()` method extracts a part of a string.

Returns the extracted part in a new string.

Does not change the original string.

`str.slice(startIndex, endIndex);`

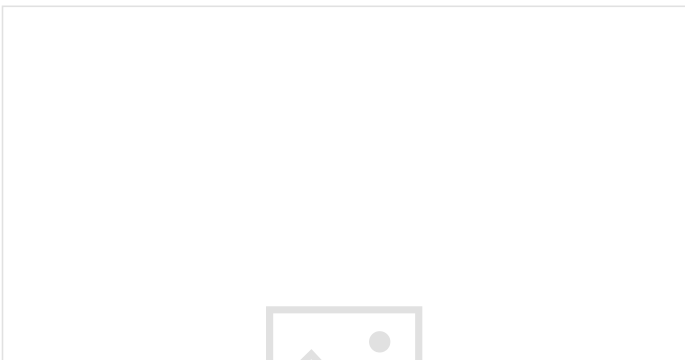
`str.slice(-1)` -> gives the last letter

- To get a substring b pattern matching:



Method Chaining

Do while loop





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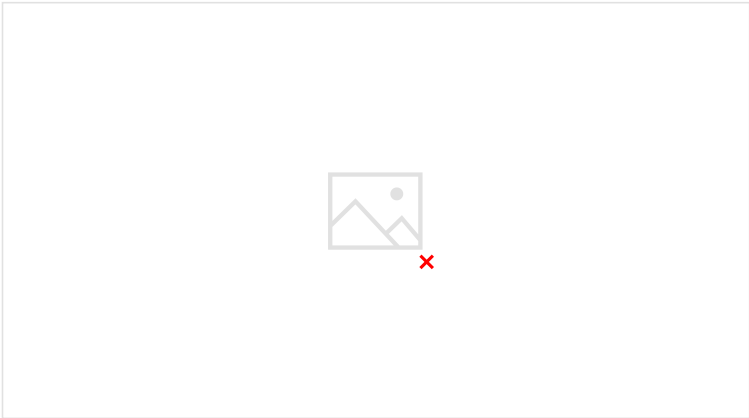
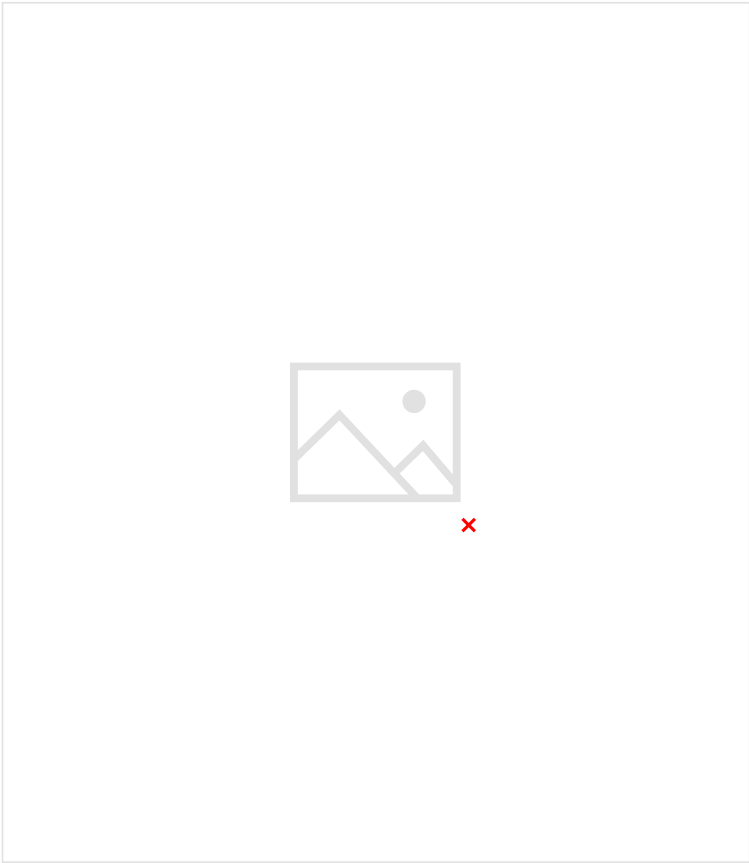
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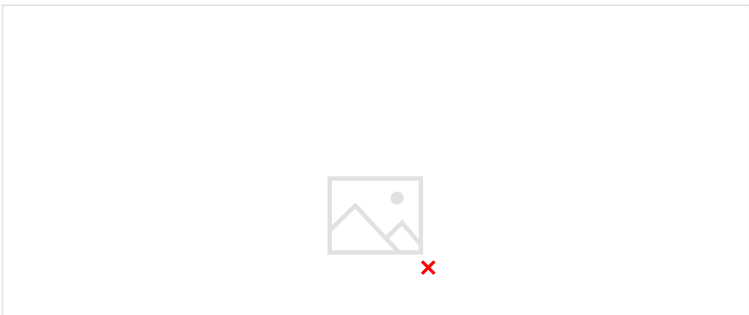
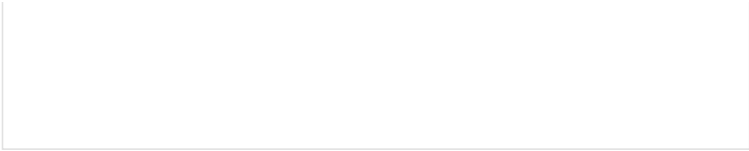


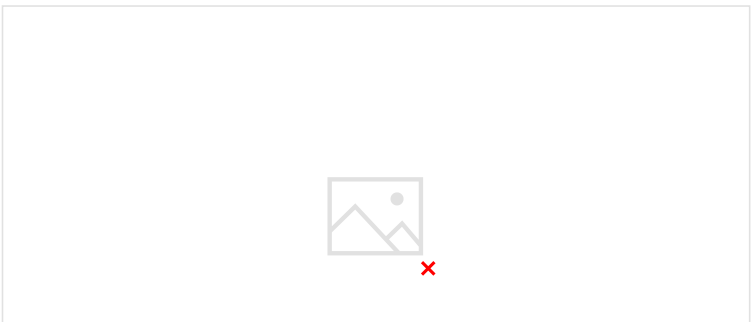
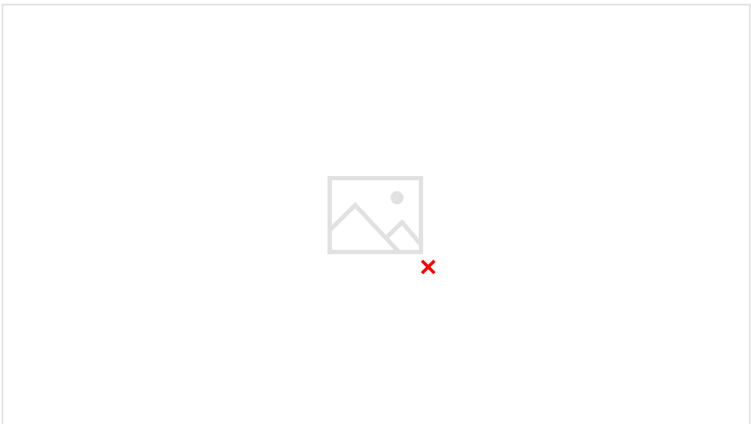
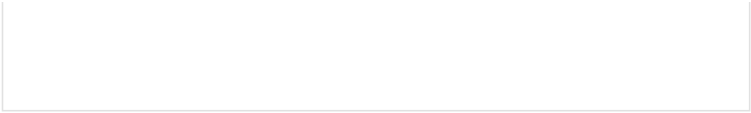
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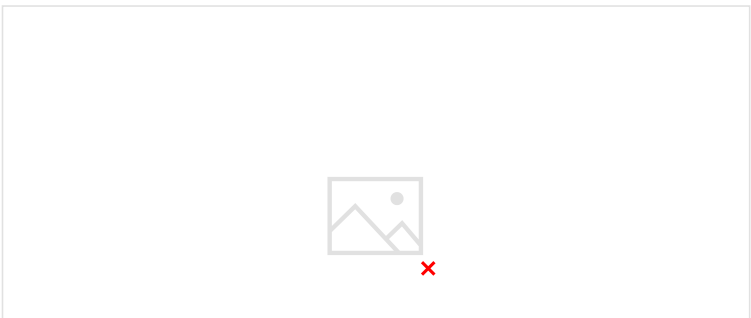


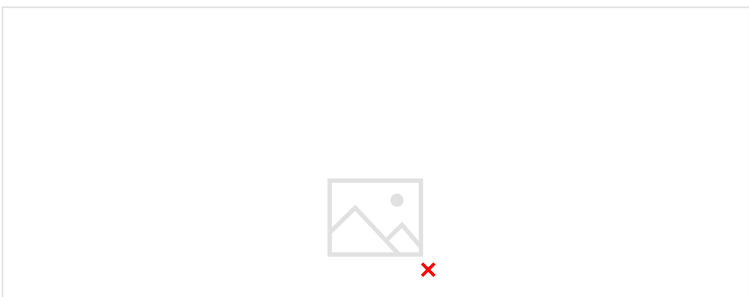
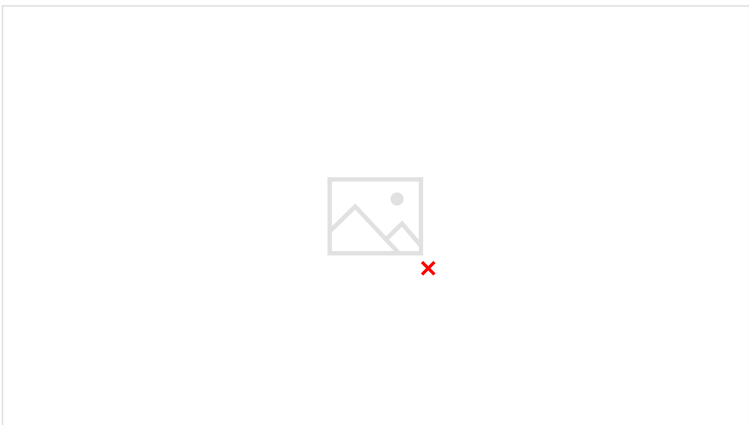
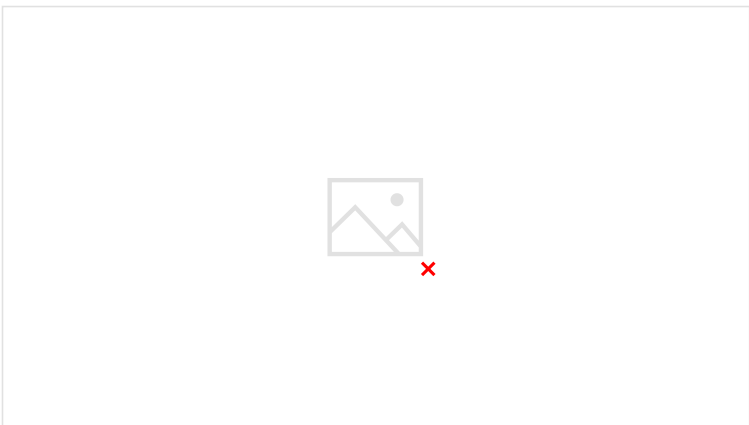
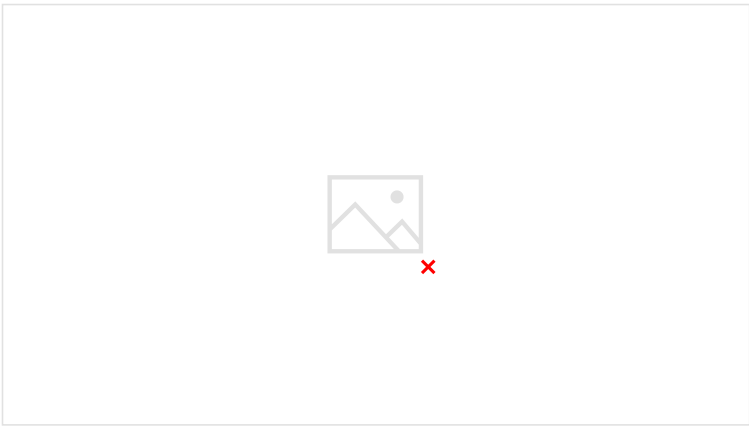


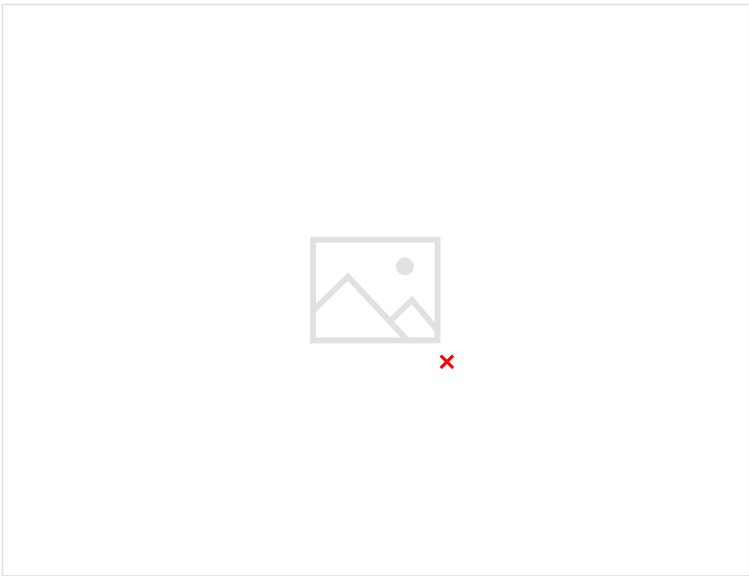














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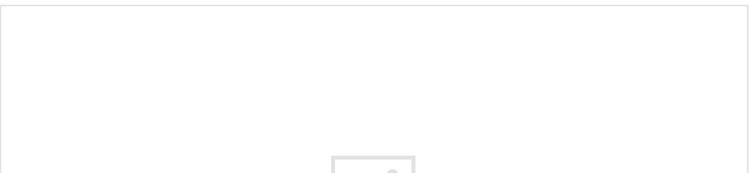
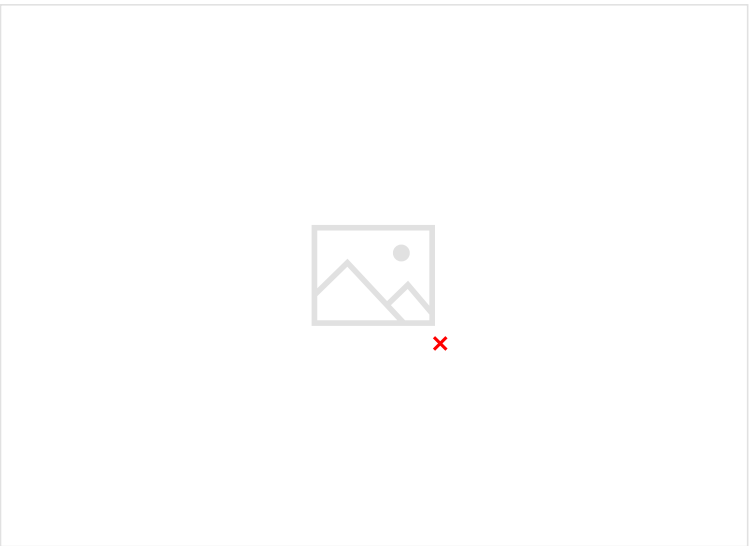
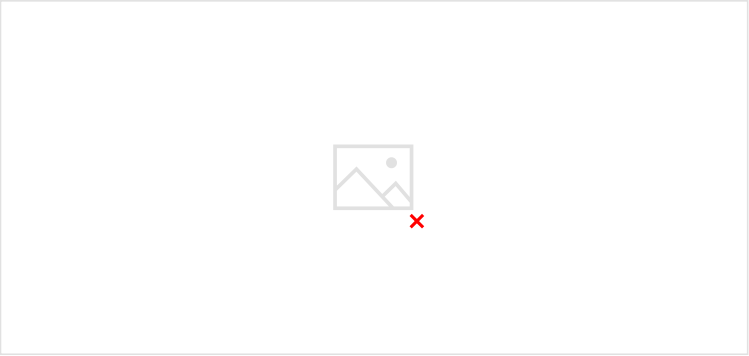


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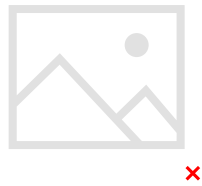
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Dynamically change css custom properties using name and data-size attributes

```
const option2 = new Option(unit, unit);
```

...is using the Option constructor, which is a built-in browser API for creating `<option>` elements for `<select>` dropdowns in HTML.

js

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Edit

new Option(text, value)

Parameter Description

text The text that will be visible in the dropdown.

value The value that will be submitted or retrieved via JS when selected.

So:

js

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new Option("kilometer", "kilometer")

Creates:

html

Copy

Edit

<option value="kilometer">kilometer</option>

You can then add it to a <select> like this:

js

Copy

Edit

unitSelectInput.appendChild(new Option("kilometer", "kilometer"));

To copy something by clicking button

navigator.clipboard.writeText(text) is the modern, secure way to copy text.

